MQM

Multi-Quantum Mode

Superposition Algorithm for High-Bandwidth Data Transmission

Inventor: Sungtae Park (scwpark)
Co-workers: Hyunuk Pak (iamgilchi@naver.com) · EunHwan Park (dmsghks0421@gmail.com)
Affiliation: Independent Researcher
Contact: scwpark@naver.com · scwpark@daum.net
Deleted (scwpark@lge.com, scwpark@hananet.net)
https://cafe.daum.net/scwpark ·
https://github.com/scwpark/MQM-PAM_OFDM_Phase-superposition_3D-mapping

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<1> Title & Author Information

- **Title:** Multi-Quantum Mode (MQM) Superposition Algorithm for High-Bandwidth Memory (HBM) and all electronics, using Frequency, PAM and Phase-based Mapping (3D Mapping data)
- **Author:** Sungtae Park (Independent Researcher)
- **Co-workers:** Hyunuk Pak (iamgilchi@naver.com), EunHwan Park (dmsghks0421@gmail.com)
- Contact: scwpark@naver.com, scwpark@daum.net
- **Core Concept:** 3D mapping of data transmission (Amplitude + Frequency + Phase) enabling massive data transfer in 1 cycle — applicable to CPU/GPU/NPU internals, RAM, HBM, and all data channels.

전송 data Symbol = (Frequency , Amplitude , Phase)

Core formula : $S(t)=A\sin(2\pi ft+\phi)$

- **A** → Amplitude (전압)
- **f** → Frequency (주파수)
- **φ** → Phase (위상)
- **t** → time (주기)

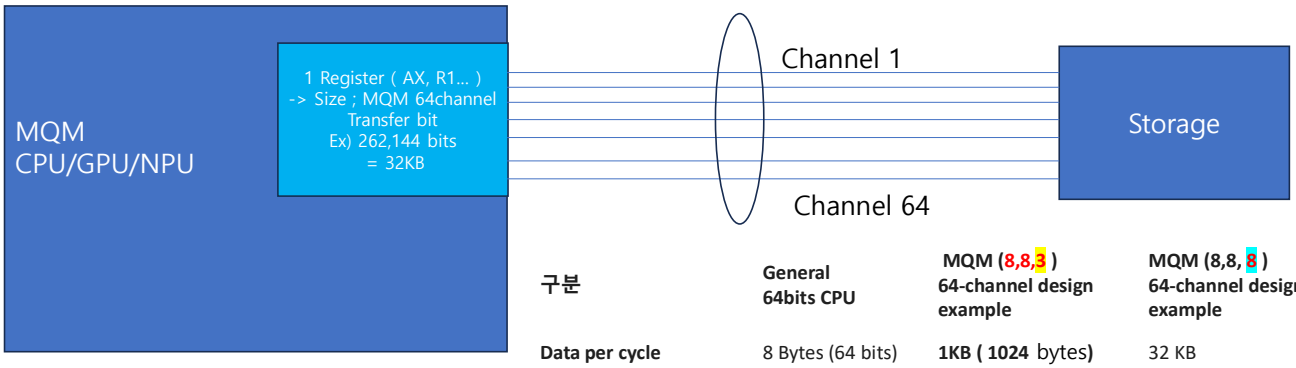
1-cycle data ⇒ 3D-mapping ⇒ High-speed transfer

"MQM employs 3-dimensional mapping — synthesizing Amplitude, Frequency, and Phase — to transmit massive data volumes in a single cycle. The target application scope encompasses all data channels: CPU/GPU/NPU cores, their internal buses, RAM, HBM, and every interconnect layer in modern computing systems."

▷ MQM structure
Data = (Frequency (n), Amplitude(m), Phase(p)) 3 dimensional data mapping

1) 1channels : (n bit, m bit, p bit) n=8, m =8, p=3 bit(8)면 =➔ **16 * 8 bit (=128) transfer (1cycle 128 bit transfer)**
2) 1channels : (n bit, m bit, p bit) n=8, m =8, p=8 bit(256)면 ➔ **16 * 256 bit(=4096) transfer(1cycle 4096 bit transfer)**

64 channels
1) 1channels : (n bit, m bit, p bit) n=8, m =8, p=3 bit(8)면 =➔ **16 * 8(= 128bit=16Bytes transfer (1cycle 16 bytes))**
2) 64 channels : 16 * 64 = 1 KB = Register size



구분	General 64bits CPU	MQM (8,8,3) 64-channel design example	MQM (8,8, 8) 64-channel design example
Data per cycle	8 Bytes (64 bits)	**1KB (1024** bytes)	32 KB
Transfer efficiency	x1	**128 x faster**	4096 x faster
analogy	1-lane country road	**128-lane expressway**	4096 –lane superhighway

<2> Introduction

MQM (Multi-Quantum Mode) is a novel ultra-high-speed parallel transmission protocol that solves the heat dissipation problem inherent in conventional linear data transmission methods, by incorporating a scaling factor into multi-dimensional wave superposition.
Unlike traditional linear data transmission, MQM encodes data across three independent physical dimensions simultaneously — Frequency (F), Amplitude (A), and Phase (P) — and superimposes them into a single composite waveform for transmission. This fundamentally departs from the von Neumann bit-by-bit paradigm.
Note: *This algorithm is released as prior art in the public domain. No patent is intended or sought by the author. Please do not attempt to patent this work.*

<3> Core Design Principles

1) Bit-Frequency/Voltage Mapping: n bits of data are transmitted simultaneously on a single channel. Each bit carries an independent frequency and voltage value.
2) Bit Frequency Formula: **F(n) = Base + 2^(n−1) × Scaling Factor**
where: n ≥ 1 (natural number), Base = center frequency/amplitude,
Scaling Factor = amplification/reduction value for hardware recognition.
This formula guarantees perfect data integrity even when multiple bits are superimposed, because each bit occupies a unique, non-overlapping value in the composite signal.

<4> Detailed Technical Specifications

[1] OFDM-Analogous MQM (Frequency-Based)

1. The Core Logic

: Transmit n bits simultaneously on 1 channel → **32 channels × n bits = total bits sent in 1 cycle**

1) Bit-Frequency Mapping: Each bit (1 ~ n) is assigned a unique, independent frequency.

2) Frequency spacing formula: F(n) = Base + 2^(n−1) × Scaling Factor [n ≥ 1]

▶ MQM Frequency Table (Example: Hz units)

(Units may be Hz, kHz, MHz, GHz as appropriate. Composite values must never collide with individual bit values.)

Bit (n)	Frequency (Hz)	Bit (n)	Frequency (Hz)
1	1	9	256
2	2	10	512
3	4	11	1,024
4	8	12	2,048
5	16	13	4,096
6	32	14	8,192
7	64	15	16,384
8	128	16	32,768

GHz example (Base = 1 GHz, Scaling Factor = 0.001):

Bit (n)	Frequency (GHz) — Base=1, SF=0.001
1	1.001
2	1.002

Bit (n)	Frequency (GHz) — Base=1, SF=0.001
3	1.004
4	1.008
5	1.016
6	1.032
7	1.064
8	1.128

Physical advantage: Prevents frequency explosion and heat buildup when transmitting 8-bit data. Maximum frequency (all 1s): 255 Hz. Maximum voltage: ~4.47 V.

2. Binary-to-Frequency (Hz) Mapping Table

Each bit position has a unique base frequency. The final transmission frequency is the sum of all active-bit frequencies (superposition).

Binary Value	Active Bits	MQM Calc (Hz)	Notes
0	None	0	No signal (Ground)
1	1st bit	1	Base frequency
10	2nd bit	2	2nd channel
11	1st + 2nd	3	First superposition
100	3rd bit	4	State bit (+1) applied
101	1st + 3rd	5	Composite waveform 1
110	2nd + 3rd	6	Composite waveform 2
111	1st+2nd+3rd	7	Full bit superposition
1000	4th bit	8	State bit (+1) re-applied

3. Modulation & Demodulation

- Superposition: For n=32 bits, only the frequencies corresponding to '1' bits are selected and superimposed into a single composite waveform for simultaneous single-line transmission.
- Integrity: Micro-spacing between frequencies acts as a natural guard band, enabling accurate data recovery even in high-noise environments — more reliable than quantum computers.

1) Application examples:

- 00010001 (binary): Classical → 8 cycles; MQM → 1 cycle at 17 Hz
- 00000001 00010001 (binary, 16-bit): Classical → 16 cycles; MQM → 1 cycle at 257 Hz

2) 2D MQM (OFDM + PAM):

- 00000001 00010001: MQM → 1 cycle at 1 Hz (1st byte) × 9 mV (2nd byte) — 2D mapping
- Or: 1 mV (1st byte) × 9 Hz (2nd byte) — 2D mapping

[2] PAM-Based MQM (Amplitude-Based)

The AMPLITUDE method, similar to "[1].OFDM-Analogous MQM" , enables n bits to be transmitted on a single channel by mapping bit values accordingly.

1. PAM MQM Principle

Instead of frequency, data is encoded in the signal's magnitude (voltage). This is a precision operation assigning bit values to fine voltage levels.

1) Current PAM standards:

- PAM-2 (NRZ): 0V(0), 1V(1) → 1 bit per transmission
- PAM-4: 0V(00), 0.33V(01), 0.66V(10), 1V(11) → 2 bits per transmission

▶ **MQM Amplitude Formula: A(m) = Base + 2^(m−1) × Scaling Factor [m ≥ 1]**

▶ MQM Amplitude table

Bit (m)	Amplitude (mV, V)	Bit (m)	Amplitude (mV, V)
1	1	9	256
2	2	10	512
3	4	11	1,024
4	8	12	2,048
5	16	13	4,096
6	32	14	8,192
7	64	15	16,384
8	128	16	32,768

▶ **PAM MQM advantage:**

- Maximum voltage (8-bit all 1s): 255 mV (0.255 V) — achievable with appropriate scaling
- n-bit superposition transmission is possible while using the same power infrastructure

2. Binary-to-Amplitude Mapping Table

Binary Value	Active Bits	MQM Calc (mV)	Notes
0	None	0	No signal (Ground)
1	1st bit	1	Base amplitude
10	2nd bit	2	2nd channel
11	1st + 2nd	3	Amplitude superposition begins
100	3rd bit	4	State bit (+1) applied
101	1st + 3rd	5	Composite waveform 1
110	2nd + 3rd	6	Composite waveform 2
111	1st+2nd+3rd	7	Full superposition
1000	4th bit	8	State bit (+1) re-applied

2D Application (MQM OFDM + PAM):

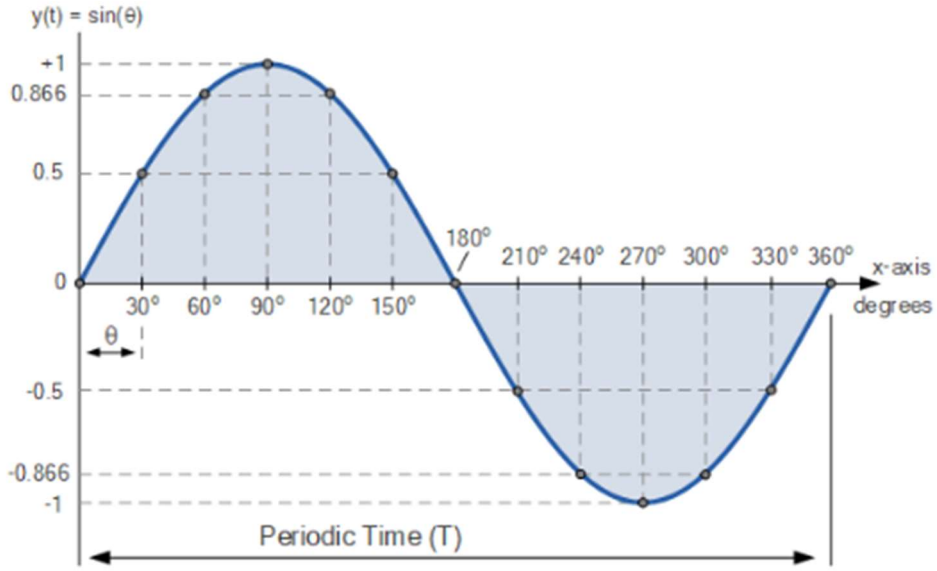
- 00000001 00010001 (binary): MQM → 1 cycle at 1 Hz (1st byte) × 9 mV (2nd byte) — 2D mapping

[3] Phase-Based MQM

1. Wave Equation

$$S(t)=A\sin(2\pi ft+\phi)$$

- **A** → Amplitude (voltage)
- **f** → Frequency
- **φ** → Phase
- **t** → time (period)



Phase encodes x bits (x is a natural number). Current 5G uses 256-QAM (8 bits). 6G research targets 16384-QAM (14 bits) and beyond.

▶ MQM Phase table – 2 bit example

Bits	Phase
00	0°
01	90°
10	180°
11	270°

▶ MQM Phase table – 3 bit example

bits	phase
000	0°
001	45°
010	90°
011	135°
100	180°
101	225°
110	270°
111	315°

2. Final MQM 3D Structure

Data = (Frequency(n), Amplitude(m), Phase(p)) — 3D data mapping

- n=8, m=8, p=3: 16 × 8 = 128 bits per cycle (16 bytes/cycle)
- n=8, m=8, p=8: 16 × 256 = 4,096 bits per cycle

▶ 14-bit Phase Mapping Design

In a 14-bit system, each data index i has a phase angle $\theta_i = i \times (360° / 2^p)$.

14 bits → 2^14 = 16,384 distinct phase states = 16,384 data values per phase slot.

With n=8 (freq), m=8 (amp), p=14 (phase): **(8+8) × 2^14 = 262,144 bits = 32 KB per cycle**

Data (i, decimal)	14-bit Binary	Phase Angle (θ_i)	Physical Meaning
0	00 0000 0000 0000	0°	Aligned with reference wave (max constructive interference)
1	00 0000 0000 0001	0.02197°	Start of micro phase shift
4096	01 0000 0000 0000	90°	Quadrature phase
8192	10 0000 0000 0000	180°	Full inversion (max destructive interference)
16383	11 1111 1111 1111	359.978°	Last data point in one cycle

Throughput Calculation:

- Base unit: 32 KB (32,768 bytes) per cycle
- At 1 GHz clock: 32 KB × 1,000,000,000 = 32.768 TB/s

⇒ A single optical cable with MQM achieves 32.768 TB/s — a true quantum jump in fiber optics!

[4] Data Compression Integration (e.g., Huffman)

Multiple encoding methods can be combined with MQM:

Data Method 1 — Direct MQM

Raw data (RAM/SSD/Flash)

→ MQM table conversion

[→ optional Fourier encoding]

→ transmit to CPU/GPU/NPU

→ [→ optional Fourier decoding]

→ MQM table decoding

→ CPU/GPU/NPU processing

Data Method 2 — Compression + MQM

Raw data

→ [Encode] Huffman/compression

→ MQM table conversion [→ optional Fourier encoding/decoding]

→ MQM decode → [Decode] Huffman decompression

→ CPU/GPU/NPU

Data Method 3 — Storage + MQM

All data stored on SSD with Huffman compression

→ [MQM table conversion → optional Fourier]

→ [optional Fourier → MQM decode]

→ Huffman decode → CPU/GPU/NPU

MQM Instruction Set Architecture (ISA) — Proposed

Mnemonic	Operands	Description	Architectural Logic
MQM.LOAD	R1, [Mem]	Load MQM Packet	Loads raw data into MQM registers via frequency/voltage mapping
MQM.SUM	R1, R2, R3	Superposition Sum	Wave superposition register addition (R1 = R2 + R3)
MQM.FFT	R1, R2	Fourier Transform	Real-time conversion of time-domain data to frequency domain
MQM.P.SHIFT8	R1, R2	Phase Shift (×8)	8-phase shift for 8× data density acceleration
MQM.P.SHIFT256	R1, R2	Phase Shift (×256)	256-phase shift for 256× data density acceleration
MQM.SYNC	R1	Sync & Guard Band	Natural guard band correction and hardware synchronization (Data Integrity)

[5] Final MQM CPU/GPU/NPU Design — Quantum Jump

▶ **MQM has two primary objectives:**

- Data Transfer: copper wire, optical fiber, wireless — all transmission media
- MQM ALU Design (CPU/GPU/NPU): Quantum jump beyond conventional CPU limitations
-

⇒ **MQM (Multi-dimensional QAM) ALU** operates on the same conceptual frontier as quantum computers, but is implementable with today's electronics.

[See companion document: 6.MQM_ALU_Wave Compute(26.03.04-03.xx)SCW_G.pdf]

Register size: ALU developers can partition the 1-cycle received data as appropriate for their architecture.

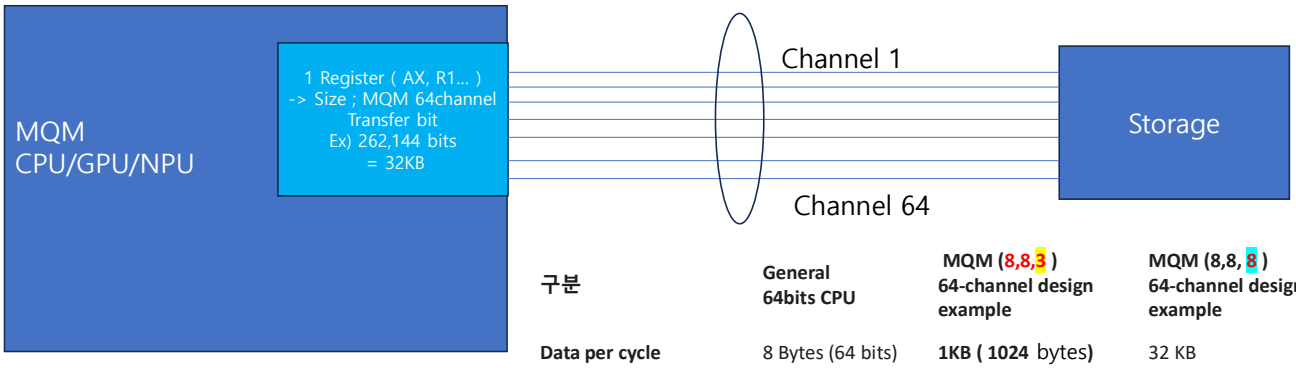
Scalability: As channel count rises (e.g., 4096 channels), ALU throughput scales dramatically.

[5.1]. MQM data transfer

▷ MQM structure
Data = (Frequency (n), Amplitude(m), Phase(p)) 3 dimensional data mapping

1) 1channels : (n bit, m bit, p bit) n=8, m =8, p=3 bit(8)면 =➔ **16 * 8 bit (=128) transfer (1cycle 128 bit transfer)**
2) 1channels : (n bit, m bit, p bit) n=8, m =8, p=8 bit(256)면 ➔ **16 * 256 bit(=4096) transfer(1cycle 4096 bit transfer)**

64 channels
1) 1channels : (n bit, m bit, p bit) n=8, m =8, p=3 bit(8)면 =➔ **16 * 8(= 128bit=16Bytes transfer (1cycle 16 bytes))**
2) 64 channels : 16 * 64 = 1 KB = Register size



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Scalability: As channel count rises (e.g., 4096 channels), ALU throughput scales dramatically.

MQM Channel Scaling : Quantum Jump Extension

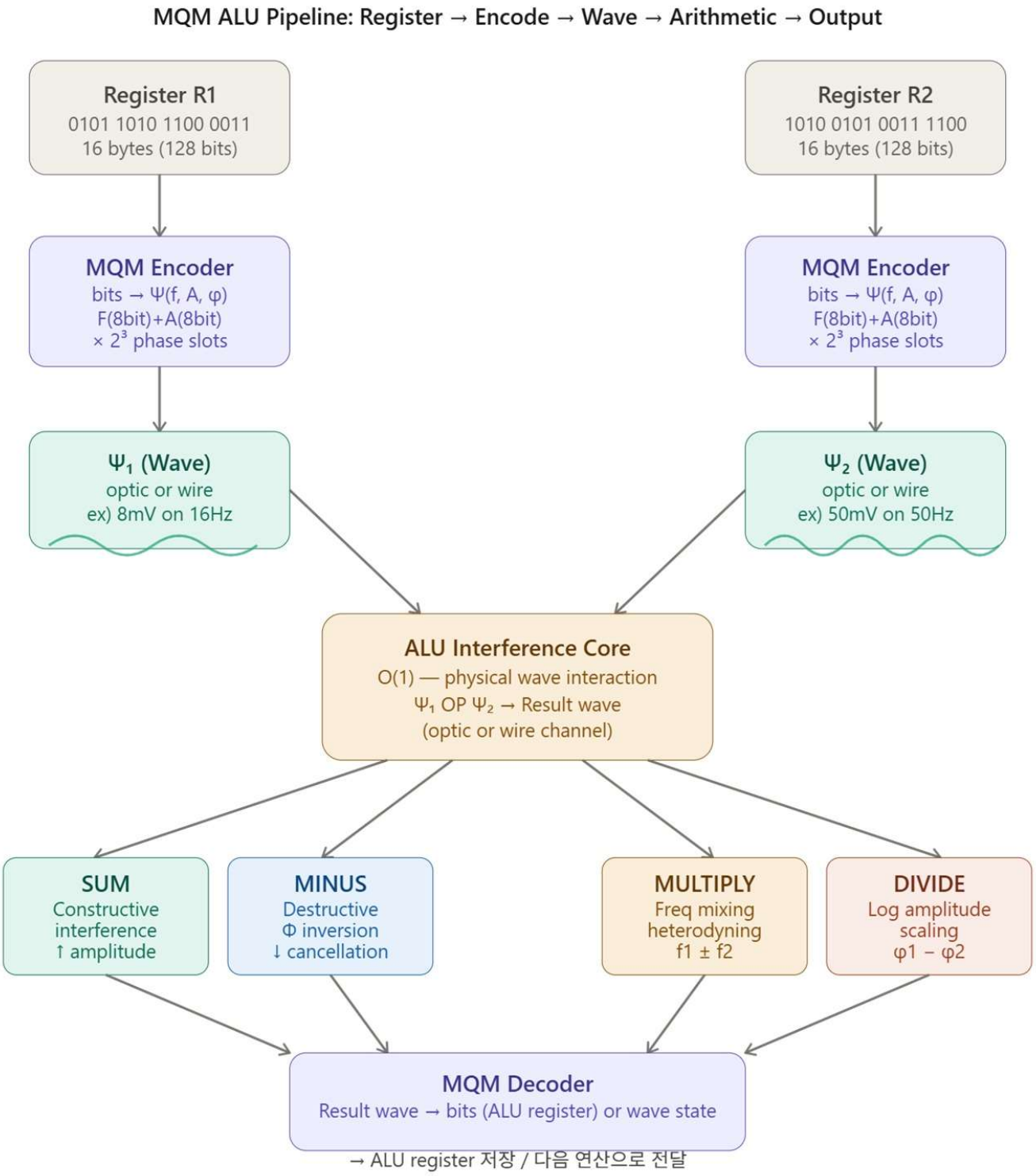
Channel 수 확장 시 성능

Channel 수	Register Size	전송효율 (vs 64bit CPU)
64 ch	32KB	4,096배
128 ch	64KB	8,192배
256 ch	128KB	16,384배
1024 ch	512KB	65,536배
4096 ch	2MB	262,144배

핵심 원리

- Channel 수 증가 = 주파수 상승 없이 성능 선형 증가
- 물리적 wire 추가 없이 MQM 인코딩으로 확장 가능
- HBM4 (2048 wire) 적용 시 즉시 구현 가능
- 🟣 4096 channel (n=8, m=8, p=8) 일 때 데이타량이 21큐비트 양자 컴퓨터 와 유사

[5.2] MQM ALU Design 2



MQM ALU — Wave-based Arithmetic Operations

Ψ(f, A) = wave state encoded by Frequency + Amplitude in a given Phase slot

1) SUM (A + B)

Constructive wave interference
두 파동의 중첩 (Superposition)

Ψ(f1, A1)
Ψ(f2, A2)
Sum_WAVE
= bigger amplitude

2) MINUS (A − B)

Destructive interference via Φ inversion
위상 반전 (Destructive Interference)

Ψ(f1, A1)
Ψ(f2, A2) + Φ(180°)
minus_WAVE = near-zero (cancellation)

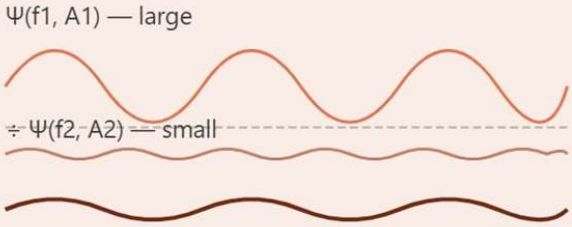
3) MULTIPLY (A × B)

Frequency mixing / heterodyning
주파수 혼합 (Frequency Mixing)

multiply_WAVE = f1±f2 product
× cross-frequency components

4) DIVIDE (A ÷ B)

Log amplitude scaling + phase subtraction
진폭 스케일링 및 위상 감산



Divide_WAVE = log(A1/A2), φ1 − φ2

Implementation Logic — O(1) Wave Processing

1) Input
Ψ1, Ψ2 via
optic or wire

→

2) Execution
Interference core
O(1) constant time

→

3) Output
bits → register or
wave → ALU reg

**"Not von Neumann's classical bit arithmetic —
but computation through physical wave interference."**

MQM Core Formula

Data Throughput = [F(n bits) + A(m bits)] × 2^p

ex) n=8, m=8, p=3 → (8+8) × 2^3 = 128 bits (16 bytes) per cycle

Phase = State (배율)	Frequency = Data	Amplitude = Data	Best: Case A)

[6] MQM Optical Communication — Quantum Jump

▶ Core Mechanism: Wave Superposition at its Extreme

Using optical fiber or silicon photonic waveguides instead of electrical conductors, MQM's 3D mapping is physically realized as follows:

▶ Optical Data Symbol = (Frequency(n bit), Amplitude(m bit), Phase(p bit))

- Frequency (n): Wavelength Division Multiplexing (WDM) — multiple optical channels superimposed. Mapping: allocate available wavelength bands (λ1, λ2 ... λn) as logical channels.
  Formula: F(n) = Base + 2^(n−1) × SF(scaling factor)
- Amplitude (m): Precision brightness control via PAM. Formula: A(m) = Base + 2^(m−1) × SF
- Phase (p): High-resolution phase control via light interference — eliminates electrical noise.
  $\theta_i = i \times (360° / 2^p)$

Example: n=8, m=8, p=14 → (8+8) × 2^14 = 262,144 bits = 32 KB per cycle

At 1 GHz: 32.768 TB/s per optical cable. If implementable inside ALUs — truly great.

▷ **Phase (p):** By precisely controlling phase using **light interference**, it completely eliminates electrical noise and projects data of **p=14 bits or higher resolution**.

⇒ **MQM phase table application:**

$$\theta i = i \, X \, \frac{360}{2^{\wedge}p}$$

$$\theta_i = i \times \frac{360^\circ}{16384} \approx i \times 0.02197265625^\circ$$

(1) *i* (data index): The actual data sequence to be transmitted

(2) 2^p: Determines how many divisions the phase is split into

(ex) p = 14 bits => 2^14 = up to 16,384 divisions

=> Therefore, if optical frequency (n = 8bit) and amplitude (m = 8bit),

then (8+8) × 2^14 = 262,144 bits = 32KB

Data (i, Decimal)	14 bit (Binary)	phase (θi)	meaning
0	00 0000 0000 0000	**0˚**	Aligned with reference wave (max constructive interference)
1	00 0000 0000 0001	**0.02197˚**	Start of micro phase shift
4096	01 0000 0000 0000	**90˚**	Quadrature phase
8192	10 0000 0000 0000	**180˚**	Full inversion (max destructive interference)
16383	11 1111 1111 1111	**359.978˚**	Last data point in one cycle

▷ Data Throughput Calculation

Transmission speed is determined by (data volume per cycle) × (clock frequency).

- **Base unit: 32KB per cycle (32,768 bytes)**
- **Assumption (system clock): Based on 1 GHz (1 billion cycles per second)**

Formula: 32KB × 1,000,000,000 cycles/sec = Result: 32.768 TB/s (Terabytes/second)

=> At this point, a "quantum jump" is occurring in the current network optical communications — a single optic cable transferring 32.768 TB/sec.

If this can also be implemented inside an ALU... that would truly be Great.

[7] Conclusion & Performance Summary

Data Symbol = (Frequency(n), Amplitude(m), Phase(p)) — all three dimensions can be used, or any two.

=> Combined method "[1].OFDM-Analogous MQM" + "[2].PAM-based MQM" (+ "[3].Phase-based MQM"): Transmits (n + m) × 2^p bits per cycle on a single channel.

- (n=m=16, p=3 bits) → 32 × 8 = **256 bits** per cycle
- (n=m=8, p=3 bits) → 16 × 8 = **128 bits** per cycle

Ex 1) Using only (Frequency 8-bit) 00000111 + (Amplitude 8-bit) 00001000

=> 7Hz, 8mV — **"2D mapping"**

Ex 2) When 8 phases (phase 3-bit) are included:

[Phase 0] (Amplitude 8-bit) 01011100 + (Frequency 8-bit) 10110011

.....

[Phase 7] (Amplitude 8-bit) 11011100 + (Frequency 8-bit) 11110011

=> 8 phases × 16-bit transmission — **"3D mapping"**

(Which bits come first or last is determined by the developer)

Concrete Application 3)

00000011 00010001 (binary) transfer method

1) Conventional method: send over 16 cycles, or encode the 16-bit value via Fourier and then decompose after transmission.

=> **MQM method:** Send in **1 cycle** as 3Hz (1st byte) + 17mV (2nd byte) — **2D mapping**.

2) MQM 16-byte information transfer: 2 bytes (phase 0), 2 bytes (phase 1) .... 2 bytes (phase 7)

→ 16 bytes transmitted in 1 cycle via 3D mapping

.....

▶ **Single-channel throughput formula: B_ch = (n + m) × 2^p bits/cycle**

- n=m=8, p=8: (8+8) × 256 = 4,096 bits/cycle
- n=m=16, p=3: (16+16) × 8 = 256 bits/cycle

▶ **Total Register Throughput (T_total)**:

T_total = B_ch × Channels (e.g., 4,096 bits × 64 channels = 262,144 bits = 32 KB)

▶ **Expected Performance (AI & Big Data):**

The following describes the real-world computational efficiency achieved through the combination of a 32KB high-capacity register architecture and MQM transmission technology.

Task	Conventional System (5GHz, Estimated)	MQM System (Target)	Improvement
LLM Inference (Token Gen)	100 tokens/sec	400,000+ tokens/sec	Memory bottleneck fully eliminated
4K Video Rendering	10 minutes	< 0.15 seconds (600s / 4000)	Real-time ultra-fast mapping computation
HBM Heat Control	High heat from fast clock	Low heat via superposition transmission	Maximum power efficiency (PPA)

Architecture Comparison:

Metric	Conventional High-Performance System	MQM-Based System (Proposed)
Clock Frequency	3.5 GHz ~ 5.0 GHz (typical ceiling)	Same frequency (or reduced for lower heat)
Register Width	64-bit (8 Bytes)	262,144-bit (32 KB)
Data per Cycle	1 data unit	4,096 data units
Processing Method	1 Line 1 Bit (serial/parallel mix)	3D Mapping (Freq + Amp + Phase)

From the conventional HBM method of 1 wire = 1 bit transmission,

MQM upgrades to 1 wire = 8(n) + 8(m) bits transmission.

=> With **1,024 wires**: **1,024 × (8+8 bits)** transmitted per cycle,

achieving **(n+m)× faster transmission efficiency** compared to conventional methods —

and simultaneously **resolving the heat dissipation problem**.

When Phase is further integrated, even **faster transmission speeds** are expected.

Key insight:

- Conventional HBM: 1,024 wires × 1 bit = **1,024 bits/cycle**
- MQM HBM: 1,024 wires × 16 bits = **16,384 bits/cycle**
- With Phase (3D): 1,024 wires × 16 bits × 2^p = **massive throughput leap**
- Heat reduced because **more data per wire per cycle = fewer cycles needed = less switching activity**

<5> Conclusion & Waiver (결론 및 권리 포기 선언)

[English]

The scwpark MQM-PAM-OFDM-PHASE algorithm is rooted in an intuitive inspiration the author experienced in the early 1980s (a non-linear data transmission framework), and was scientifically systematized in 2026 in consultation with co-workers Hyunuk Pak and EunHwan Park, to overcome the HBM heat limits and ALU bottlenecks of modern semiconductors.

- **Origin:** The **MQM-PAM-OFDM-PHASE** algorithm originated from an intuitive inspiration the author experienced in the early 1980s — a vision of transmission technology and ALU architecture observed in a dream — and was scientifically systematized in 2026 to overcome the limitations of modern quantum computing and HBM data transfer.Public Release Purpose: Released as free open-source public interest technology to solve the chronic heat problems of linear transmission and drive a quantum leap in global semiconductor technology.
- Patent Waiver: The author (scwpark) explicitly waives all exclusive patent rights to this technology and hopes it will be freely used for the benefit of humanity.
- Anti-Monopoly: This document constitutes prior art. The author strongly opposes any third party attempting to patent this logic to restrict its free public use.

[Korean]

본 **scwpark** MQM-PAM-OFDM-PHASE 알고리즘은 1980년대 초 저자가 경험한 직관적 영감(비선형적 데이터 전송 체계)을 바탕으로 하며, 현대 반도체의 HBM 발열 한계, ALU한계를 극복하기 위해 2026년 Hyunuk Pak , EunHwan Park(dmsghks0421@gmail.com)과 상의하여 과학적으로 체계화되었습니다

기술의 기원: 본 MQM-PAM-OFDM-PHASE 알고리즘은 1980년대 초 저자의 직관적 영감(꿈속에서 본 외 x인들이 만드는 전송 기술과 ALU)에서 시작되어, 현대의 양자 컴퓨팅 및 HBM 전송 한계를 극복하기 위해 2026년 과학적으로 체계화되었습니다.

공개 목적: 기존 선형 전송 방식의 고질적인 발열 문제를 해결하고, 전 세계 반도체 기술의 비약적인 발전을 돕기 위해 오픈 소스 공익용으로 공개합니다.

특허권 포기 선언: 저자(**scwpark**)는 본 기술에 대한 모든 독점적 특허권을 명시적으로 포기하며, 이 기술이 인류를 위해 자유롭게 사용되기를 희망합니다.

타인 특허 등록 금지: 본 문서는 '선행 기술(Prior Art)'로서의 지위를 가지며, 본 로직을 이용해 제3자가 특허를 취득하여 기술의 자유로운 사용을 제한하는 것에 강력히 반대합니다.

-- -- 이건 80년대 초부터 꿈속에서 참 희한 꿈이였는데

(요즘 양자 컴퓨터 하길레.. 꿈을 과학적으로 기술해서 양자 컴퓨터 보다 빠르게..) 아들과 상의하여 작성

80, 90년대부터 내가 떠벌 떠벌 말하고 다녔는데.. 아직 HBM등 전송 기술에.. 완벽히 안들어가서

내가 작성해보네요..

특허생각은 없음...(부디 이글 보고 특허로 하지 말기합시다..)

⚖️ **Intellectual Property & Citation Policy**

본 MQM 기술은 전 세계 반도체 및 컴퓨팅 생태계의 발전을 위해 **'선행 기술(Prior Art)'**로서 오픈소스로 공개되었습니다. 저자(scwpark)는 핵심 알고리즘에 대한 모든 독점적 특허권을 포기합니다. 다만, 기술의 올바른 확산과 지적 정직성을 위해 다음 사항을 의무화합니다.

[English]

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[Korean]

출처 명기 의무화 정책:

1. **출처 명기 의무:** 본 MQM 알고리즘을 기초 기술로 활용하는 제3자의 모든 응용 특허 출원, 논문, 또는 상업적 구현물은 반드시 원천 기술인 **MQM 알고리즘**과 본 문서의 **Zenodo DOI**를 선행 기술로 명시적으로 인용해야 합니다.
2. **독점 금지:** 그 어떤 개인이나 단체도 MQM 3D 매핑 로직 자체에 대해 독점적 권리를 주장할 수 없습니다. 모든 특허 청구 범위는 원천 기술의 공공적 사용을 제한하지 않는 범위 내에서, 추가적인 독창적 구현 방식에만 국한되어야 합니다.

🧑‍🔬 Author & Credits

- **Inventor:** Sungtae Park (scwpark)
- **Co-workers:** Hyunuk Pak, EunHwan Park
- **Community:** https://cafe.daum.net/scwpark
- **Zenodo DOI(Latest):** https://doi.org/10.5281/zenodo.18930827
- **Zenodo DOI:** https://doi.org/10.5281/zenodo.18924211

☕ Buy Me a Coffee

MQM is free and open-source forever — no patent, no fee.

If MQM helped you in any way, even a single cup of coffee keeps this old doctor-inventor going! 🦋

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"I had this dream in the 1980s.
Forty years later, I'm giving it to the world for free.
A coffee would be nice though." ☕😄